

# Question 4 – Classes and Objects

Date: 24/07/2026

## Objective / Aim

To demonstrate classes, objects, constructors, and method overloading using an Animal class in Java.

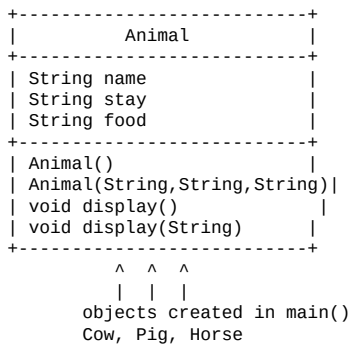
## Algorithm / Logic

1. Create the Animal class with name, stay, and food attributes.
2. Define a default constructor with Unknown values.
3. Define a parameterized constructor to initialize animal details.
4. Define display() to print the animal information.
5. Overload display() with a sound parameter.
6. Create objects for Cow, Pig, and Horse using the parameterized constructor.
7. Call display(sound) for each object.

## Steps for Execution

1. Create the Java project and add the Animal and ClassesAndOBJects classes.
2. Compile and run ClassesAndOBJects.java.
3. Create Cow, Pig, and Horse objects.
4. Pass the corresponding sound to the overloaded display() method.
5. Observe the animal details and sounds.

## UML Class Diagram



## Source Code

```
class Animal {
    String name;
    String stay;
    String food;

    Animal() {
        name = "Unknown";
        stay = "Unknown";
        food = "Unknown";
    }

    Animal(String name, String stay, String food) {
        this.name = name;
        this.stay = stay;
        this.food = food;
    }

    void display() {
        System.out.println("Animal : " + name);
        System.out.println("Stay : " + stay);
        System.out.println("Food : " + food);
    }

    void display(String sound) {
        display();
        System.out.println("Sound : " + sound);
        System.out.println();
    }
}

public class ClassesAndOBJects {
    public static void main(String[] args) {
        Animal cow = new Animal("Cow", "Farm", "Grass");
        Animal pig = new Animal("Pig", "Pigsty", "Vegetables");
        Animal horse = new Animal("Horse", "Stable", "Hay");

        cow.display("Moo");
        pig.display("Oink");
        horse.display("Neigh");
    }
}
```

## Output

Animal : Cow  
Stay : Farm  
Food : Grass  
Sound : Moo

Animal : Pig  
Stay : Pigsty  
Food : Vegetables  
Sound : Oink

Animal : Horse  
Stay : Stable  
Food : Hay  
Sound : Neigh